

All solutions should be well justified. The total amount of points is 90. Use the departments papers only. Begin the solution of each assignment **on top of a new sheet**. Do **not** use **red** pen or pencil. You are allowed to use a pocket calculator.

Write your identifier on each sheet.

1. Give the definition of a sigma-algebra $\mathcal{F}$ on some state space Ω . Give an example of a sigma-algebra which has **exactly 5 elements**, or prove that such example cannot exist. (10p)

2. You play a certain game every day during a week, starting from Monday. On the first day of the week your chance of winning is $1/10$, on the second $2/10$, and so on; on Sunday your chance of winning is $7/10$. What is the probability of winning on a randomly picked day (i.e., you choose any day of the week with equal probability)? (10p)

Suppose that I know that you did *not* win, but I do not know which day of the week it was. What is the probability that the day was Wednesday?

3. Let $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$. The variance-covariance matrix of $\mathbf{X}$ is given by $\text{Var}(\mathbf{X}) = \begin{pmatrix} 2 & -1 \\ a & a+2 \end{pmatrix}$. What should be the value of a to make this the appropriate variance-covariance matrix? (10p)

Let $\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} = \mathbf{A}\mathbf{X}$ where $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$. Find $\text{Cov}(Y_1, Y_2)$.

4. Suppose (X, Y) are jointly continuous with the density function $f_{X,Y}(x, y) = c$ for some $c \in \mathbb{R}$, when both $0 \leq x \leq 2$ and $0 \leq y \leq 2$, and $f_{X,Y}(x, y) = 0$ otherwise. (20p)

What is the value of c ? Are X and Y independent or not? Justify your answer.

Compute the density of the random variable $Z = XY$.

5. Let the joint density of random variables X, Y be given by

$$f_{X,Y}(x, y) = c \left(\frac{x}{y} + \frac{y}{x} \right), \quad x, y \in [1, 2]$$

and $f = 0$ for the other values of x, y .

(a) Determine the value of constant c for which $f_{X,Y}$ is, indeed, a density. (10p)

(b) Compute $E(Y \mid X = x)$. (10p)

6. Formulate the Strong Law of Large numbers for a sequence of independent identically distributed random variables $X_1, X_2, \dots$ (20p)

Prove it in the case $\mathbb{E}X_i = 0$ and $\mathbb{E}[X_i^k] < \infty$ for $k = 2, 3, 4$. You may use Borel-Cantelli lemma without a proof, if you need it.

Good luck!

Solutions

1. Definition: see lecture notes/textbooks. As $\mathcal{F}$ must contain A^c together with each set A and $A \neq A^c$ for all A (why?), the number of elements must be even (if finite). Hence no such $\mathcal{F}$ exists.
2. Let $W = \{\text{you win}\}$, and $D_i = \{\text{you play on day } i\}$, $i = 1, 2, \dots, 7$. Then $\mathbb{P}(W \mid D_i) = i/10$ and $\mathbb{P}(W) = \sum_{i=1}^7 \mathbb{P}(W \mid D_i) \mathbb{P}(D_i) = \sum_{i=1}^7 \frac{i}{10} \frac{1}{7} = \frac{2}{5}$ by the law of total probability (and $\mathbb{P}(W^c) = 1 - \mathbb{P}(W) = \frac{3}{5}$).

By the Bayes theorem, $\mathbb{P}(D_3 \mid W^c) = \frac{\mathbb{P}(W^c \mid D_3) \mathbb{P}(D_3)}{\mathbb{P}(W^c)} = \frac{(1-3/10) \cdot 1/7}{3/5} = \frac{1}{6}$.

3. By symmetry, $\mathbf{Cov}(X_1, X_2) = \mathbf{Cov}(X_2, X_1)$, hence $a = -1$, so $\mathbf{Var}(X_1) = 2$, $\mathbf{Var}(X_2) = 1$, $\mathbf{Cov}(X_1, X_2) = -1$. We have $Y_1 = X_1 + 2X_2$, $Y_2 = 3X_2$, so $\mathbf{Cov}(Y_1, Y_2) = \mathbf{Cov}(X_1 + 2X_2, 3X_2) = 3 \cdot \mathbf{Cov}(X_1, X_2) + 2 \cdot 3 \cdot \mathbf{Var}(X_2) = 3(-1) + 6 \cdot 1 = 3$. (Recall that $\mathbf{Cov}(Z, Z) = \mathbf{Var}(Z)$.)
4. Since the area of the square $[0, 2] \times [0, 2]$ is 4, and the density must integrate to 1, that is $\iint f_{X,Y}(x,y) dx dy = 1$, we have $c = 1/4$.

The marginal density of X is $f_X(a) = \int f_{X,Y}(x,y) dy = \int_0^2 \frac{1}{4} dy = 1/2$ when $0 \leq a \leq 2$, so $f_X(a) = 1_{0 \leq a \leq 2}$; the same is true for Y by the symmetry. Hence $f_{X,Y}(x,y) = f_X(x) \times f_Y(y)$, so the joint density is a product of the marginal densities, hence X, Y are independent.

We first start with computation of the cdf $F_Z(a) = \mathbb{P}(Z \leq a)$. Since $0 \leq XY \leq 4$, $F_Z(a) = 0$ for $a < 0$ and $F_Z(a) = 1$ for $a \geq 4$. For $a \in [0, 4]$ we have

$$\begin{aligned} \mathbb{P}(Z \leq a) &= \iint_A \frac{1}{4} dx dy = \frac{1}{4} \left[\int_0^{a/2} \left(\int_0^2 dy \right) dx + \int_{a/2}^2 \left(\int_0^{a/x} dy \right) dx \right] \\ &= \frac{1}{4} \left[\int_0^{a/2} 2 dx + \int_{a/2}^2 \frac{a}{x} dx \right] = \frac{a(1 + 2 \ln 2 - \ln a)}{4} \end{aligned}$$

where $A = \{(x,y) : 0 \leq x \leq 2, 0 \leq y \leq 2, xy \leq a\}$. Hence the density of Z is given by $f_Z(a) = \frac{d}{da} F_Z(a) = \frac{\ln(4/a)}{4}$, $a \in (0, 4)$, and 0 otherwise.

An alternative neat method is to write $\xi_1 = \ln(2/X) \in (0, \infty)$, $\xi_2 = \ln(2/Y)$ and notice that $\mathbb{P}(\xi_i > a) = e^{-a}$, $a > 0$ so ξ_i 's are iid $\exp(1)$ r.v. (since X, Y are independent) with density $f(a) = e^{-a} 1_{a>0}$. By the convolution formula the pdf of $S = \xi_1 + \xi_2$ is $f_S(s) = \int f(x)f(s-x)dx = \int_0^s e^{-x} e^{-(s-x)} dx = s e^{-s}$, $s > 0$. But $S = h(Z)$ where $h(s) = \ln \frac{4}{s}$ is a monotone function, so $f_Z(a) = f_S(h(a)) \cdot |h'(a)| = \ln \frac{4}{a} e^{-\ln \frac{4}{a}} \cdot \frac{1}{a} = \frac{1}{4} \ln \frac{4}{a}$.

5. (a) We find c from $\iint f_{X,Y}(x,y) dx dy = 1$, i.e.

$$1 = \int_1^2 \left[\int_1^2 c \left(\frac{x}{y} + \frac{y}{x} \right) dy \right] dx = \int_1^2 c \left(\frac{3}{2x} + x \ln 2 \right) dx = c \cdot 3 \ln 2$$

so $c = (3 \ln 2)^{-1} = 0.48089\dots$

(b) The marginal density of X is $f_X(x) = \int f_{X,Y}(x,y) dy = c \left(\frac{3}{2x} + x \ln 2 \right)$, hence the conditional density is $f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \frac{2x^2 + 2y^2}{y(2x^2 \ln 2 + 3)}$ and thus

$$\mathbb{E}(Y \mid X = x) = \int y f_{Y|X}(y|x) dy = \int_1^2 \frac{2x^2 + 2y^2}{2x^2 \ln 2 + 3} dy = \frac{2x^2 + 14/3}{2x^2 \ln 2 + 3} = \frac{6x^2 + 14}{6x^2 \ln 2 + 9}$$

6. For the proof, please see the lecture notes, or Theorem 2.3.5 in Durrett's "Probability" book (see e.g. https://services.math.duke.edu/~rtd/PTE/PTE5_011119.pdf)